

Smart India Hackathon 2026

Technical Report

BeamLock

AI-Based Virtual Camera Tracking System for
Coarse Alignment of Mobile Free-Space Optical
Communication (FSOC) Terminals

Problem Statement ID: SIH26169

Organization: Department of Space / ISRO

Theme: Smart Automation / Space Technology

Category: Software

Team Name: Cadets

Repository: <https://github.com/MudithDShetty/SIH>

Software package: fsoc-virtual-tracker (product name: BeamLock)

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Physics-informed PAT sandbox for algorithm development and SIH evaluation.

Not a certified BER model or flight-qualified optical terminal.

Abstract

This report documents **BeamLock**, an AI-assisted virtual camera tracking system developed for SIH26169. The software closes a real-time coarse pointing–acquisition–tracking (PAT) loop inside a configurable virtual scene: moving optical beacons, an azimuth/elevation (AZ/EL) virtual camera plant with rate limits, Classical and hybrid AI detectors, constant-velocity Kalman tracking, covariance-scaled raster re-acquisition, disturbance injection, and automated performance logging. A physics-informed atmospheric channel ($Cn^2 \rightarrow r_0$, beam wander, scintillation) and Gaussian-beam link budget feed a Link Readiness score that gates an optional coarse-to-fine handoff mock (4-quadrant detector + fast steering mirror). The system implements the problem statement’s expected solution capabilities, exposes dual *Physics* / *PS Compliance* runtime profiles, and provides an offline MP4 Benchmark-2 path with PTZ bypass for SIH evaluation Stage 2. Verified matched-seed Classical vs AI results, acquisition Monte Carlo against an ESTOL-inspired 60s bound, and representative HUD captures are reported with honest limitations (notably slew-lag-limited mean pixel error in the physics closed loop).

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1 Problem Understanding

1.1 Operational context

Free-Space Optical Communication (FSOC) can deliver gigabit-to-terabit class links using narrow laser beams in unlicensed spectrum with high immunity to radio-frequency interference. On mobile platforms (UAVs, aircraft, LEO terminals), the beam must remain aligned through a PAT process that typically comprises:

1. **Coarse alignment:** locate the remote beacon and keep it inside the acquisition camera field of view (FOV);
2. **Fine alignment:** null residual pointing error with a high-bandwidth mechanism (e.g. fast steering mirror / FSM and four-quadrant detector / 4-QD) before communications beam lock.

Hardware PAT benches require costly cameras, gimbals, and optics, which slows algorithm iteration. SIH26169 therefore asks for a **software virtual camera** in which coarse alignment algorithms can be developed, disturbed, measured, and compared—including AI assistance—without specialized hardware.

1.2 Official problem decomposition

Before fine pointing can engage, coarse alignment must:

- observe the environment through a virtual camera viewport;
- acquire and detect the remote beacon under clutter and noise;
- estimate beacon position and motion;
- continuously reposition the camera (pan-tilt class AZ/EL control).

The SIH Parameters table further specifies suggested screen/camera sizes, FOV, motion classes (including figure-of-eight), named noise and weather grades, and performance targets (acquisition time, tracking error, lock retention, FPS). The Evaluation table weights live demonstration, Benchmark Performance-1 (scenario logs), Benchmark Performance-2 (judge MP4 at ~30 fps with PTZ bypass), and technical Q&A.

1.3 Expected solution mapping

BeamLock maps each expected solution bullet to an implemented capability:

Expected capability	BeamLock realization
Virtual camera + scene	<code>scene/</code> + <code>scene/camera.py</code> AZ/EL plant
Beacon detection	Classical centroid; YOLOv8n; Hybrid fusion
Tracking / estimation	Constant-velocity Kalman (LOCKED/COASTING/LOST)
Camera repositioning	Closed-loop slew toward Kalman estimate (not GT)
Disturbances	Turbulence, vibration, sensor noise, weather grades
AI assistance	Synthetic training + live hybrid path
Performance logging	CSV + run-sheet JSON + Streamlit/PDF

1.4 Scope and non-goals

BeamLock is a real-time PAT *sandbox* (~ 60 fps class on a laptop). It is **not** a certified bit-error-rate (BER) / outage calculator, split-step phase-screen wave-optics propagator, or flight-qualified optical terminal software. Physics models follow Andrews & Phillips order-of-magnitude horizontal / UAV-class path intuition and are calibrated for demo dynamics, not flight certification.

2 System Architecture

2.1 Closed-loop overview

Figure 1 shows the end-to-end architecture. Operator configuration drives a virtual world; disturbances and link physics degrade the camera view; detection and tracking produce camera drive commands on a control bus; outputs provide live HUD metrics and archival logs. A feedback path returns AZ/EL drive (coarse, re-acquisition, or fine mock) to the virtual camera plant.

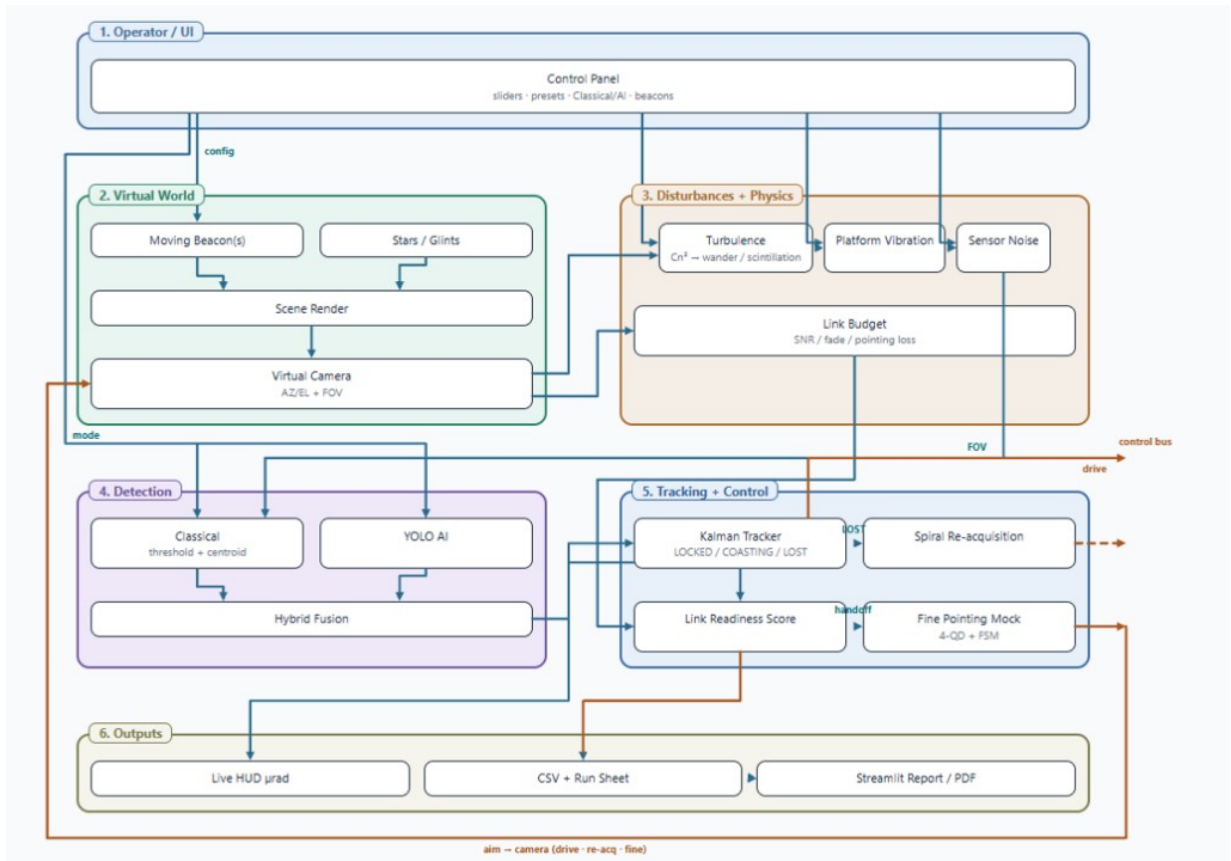


Figure 1: BeamLock real-time PAT architecture. Forward path: scene $\rightarrow$ disturbances/physics $\rightarrow$ detection $\rightarrow$ tracking/control $\rightarrow$ outputs. Feedback path: Kalman / re-acquisition / fine mock drives the virtual camera AZ/EL plant. (*Note:* the live default re-acquisition pattern is **rastrer**; spiral generation remains in code for literature comparison.)

2.2 Per-frame control law

Each simulation frame executes:

1. Advance beacon kinematics and render scene (with clutter if enabled).

2. Apply atmospheric wander/scintillation appearance coupling, platform vibration, and sensor noise to the FOV crop.
3. Compute link budget quantities (P_{rx} , SNR, pointing loss) and Link Readiness $S \in [0, 1]$.
4. Detect beacon(s) via Classical *or* Hybrid AI path (mutually exclusive live mode).
5. Update Kalman tracker; emit state `LOCKED` / `COASTING` / `LOST`.
6. Select plant command:
 - if `LOST`: raster re-acquisition waypoints with distance-scaled slew;
 - else if Fine stage active: mock 4-QD residual nulling;
 - else: coarse AZ/EL slew toward the Kalman estimate.
7. Log metrics and draw HUD (μ rad-primary angular error, lock state, readiness, FPS).

2.3 Angular geometry

Default physics acquisition FOV is scaled to an ESA IZN-1 class ~ 2.5 mrad horizontal FOV:

$$\text{IFOV} = \frac{\text{FOV}_H}{N_x} = \frac{2500 \mu\text{rad}}{400 \text{ px}} = 6.25 \mu\text{rad/px}. \quad (1)$$

Pixel error maps to angular error by $\theta_{\mu\text{rad}} = e_{\text{px}} \cdot \text{IFOV}$. The AZ/EL plant maintains boresight in scene coordinates with rate and acceleration limits (UI slew in px/s, equivalently $\mu\text{rad/s}$ via IFOV). During `LOST` search, slew multipliers increase with distance to the current waypoint (capped by `MAX_ACQ_SLEW_MULTIPLIER`).

3 Description of Software Modules

3.1 Module inventory

Table 1: Primary software modules and responsibilities

Module / path	Responsibility
ui/	Sliders, Calm/UAV/Stress presets, Classical/AI toggle, beacon count, trajectory selection (linear, circular, random, figure-8), noise type and weather keys.
scene/	Sky background, moving beacon(s), optional star/glint clutter, Gaussian or square beacon rendering.
scene/camera.py	FOV crop, AZ/EL plant with rate limits, true IFOV ($\mu\text{rad}/\text{px}$).
disturbance/	Warp residual, vibration AR(1), sensor-noise coupling into FOV.
physics/atmosphere.py	$C_n^2 \rightarrow r_0$, Rytov variance, beam wander, scintillation sampling.
physics/link_budget.py	Gaussian-beam P_{rx} , SNR, fade margin, pointing loss.
detector/	Classical threshold centroid; YOLOv8n; Hybrid fusion; preprocess.
tracker/kalman_tracker.py	Constant-velocity KF; LOCKED / COASTING / LOST.
control/reacquisition.py	Variance-scaled raster search (default activate).
control/fine_pointing.py	Readiness-gated 4-QD + FSM mock.
metrics/	Link readiness, CSV schema, session metrics, logger.
report.py	Streamlit analysis + PDF export.
profiles.py	physics (default) vs ps (SIH table sizes/FOV).
scripts/	Comparison matrix, Monte Carlo, Pd/Pfa, ablation, demo video, Benchmark-2, training data, EXE build.
main.py	Live closed-loop orchestration.

3.2 Runtime profiles

- **Physics (default):** 1280×720 scene, 400×300 FOV crop, $2500 \mu\text{rad}$ horizontal FOV class (IFOV = $6.25 \mu\text{rad}/\text{px}$), Gaussian beacon appearance suited to optical demo intuition.
- **PS compliance (--profile ps):** 2000×2000 screen, 640×480 camera, $4^\circ \times 3^\circ$ FOV, $\sim 5^\circ/\text{s}$ slew class, square $\sim 10 \times 10$ beacon aligned to the Parameters table.

Physics mode remains the default so atmospheric channel demos are not broken by degree-scale FOV geometry; PS mode is used when judges evaluate against the Parameters table.

3.3 Atmospheric channel (technical)

UI turbulence strength $s \in [0, 10]$ maps log-uniform onto $[C_{n,\min}^2, C_{n,\max}^2]$ with $C_{n,\min}^2 = 10^{-17}$ and $C_{n,\max}^2 = 10^{-12} \text{ m}^{-2/3}$:

$$C_n^2(s) = 10^{\log_{10} C_{n,\min}^2 + (\log_{10} C_{n,\max}^2 - \log_{10} C_{n,\min}^2)(s/10)} \quad (s > 0). \quad (2)$$

Default optical assumptions: $\lambda = 976 \text{ nm}$ (acquisition-beacon class), path $L = 8 \text{ km}$, receive aperture $D = 0.08 \text{ m}$. Fried parameter (plane-wave form in code):

$$r_0 = (0.423 k^2 C_n^2 L)^{-3/5}, \quad k = 2\pi/\lambda. \quad (3)$$

Rytov variance $\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$ drives scintillation regime selection (lognormal weak; gamma-gamma style product in moderate/strong, clamped for real-time stability). Beam wander uses an angle-of-arrival scale $\sim 0.5\lambda/r_0$ blended with a C_n^2 -dependent floor and AR(1) temporal memory ($\alpha \approx 0.85^{60\Delta t}$). Wander offsets the drawn beacon; scintillation and pointing loss modulate intensity before detection.

3.4 Link budget and Link Readiness

Received power follows a directed FSO product form:

$$P_{rx} = P_{tx} \eta_{tx} \eta_{rx} G_{\text{geo}} T_{\text{atm}} L_{pe} I_{\text{scint}}, \quad (4)$$

with Gaussian pointing loss $L_{pe} = \exp(-2(\theta/\theta_{\text{div}})^2)$. SNR and a 10 dB handoff floor define fade margin for readiness (not a claimed BER threshold). Physics-path Link Readiness:

$$S = \text{clamp}(0.45 s_{\text{SNR}} + 0.25 L_{pe} + 0.15 \min(1, I) + 0.15 s_{\text{lock}}, 0, 1), \quad (5)$$

with hard rule: if tracking state = LOST then $S = 0$. s_{SNR} maps roughly 10 dB $\rightarrow 0$ and 25 dB $\rightarrow 1$.

4 Tracking Methods

4.1 Classical detection

Bright-spot detection via grayscale thresholding (default threshold ≈ 200) and largest-contour centroid. Adaptive threshold mode improves robustness under elevated sensor noise and weather grades. Classical remains the transparent baseline and the preferred detector for offline Benchmark-2 when ground-truth centroid scoring is the objective.

4.2 State estimation (Kalman)

A four-state constant-velocity filter estimates $\mathbf{x} = [x, y, v_x, v_y]^\top$ in scene coordinates:

$$F = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \quad (6)$$

Process noise Q uses the standard discrete white-noise acceleration structure scaled by process-noise intensity q (default $q = 25$); measurement noise $R = \sigma_z^2 I_2$ (default $\sigma_z^2 = 4$). Initial covariance is large ($P \propto 500$) until measurements arrive.

Table 2: Tracking state machine

State	Condition
LOCKED	Valid detection accepted this frame
COASTING	1–15 consecutive misses (coast on prediction)
LOST	> 15 misses (or forced loss on detector switch / blind)

The camera is driven by the filter estimate, preserving a true closed loop (ground truth is used only for scoring and HUD truth overlays).

4.3 Re-acquisition

On LOST, `ReacquisitionController` activates a **raster** waypoint pattern (default). Step spacing scales with position covariance so search expands under uncertainty. Search center is biased by last lock position, last velocity, and time-since-loss. Slew rate increases with waypoint distance to reduce corner-trap failures observed with spiral-only search under large uncertainty. Spiral generation remains in code for literature comparison but is not the live activate path. Plausibility gating rejects false re-locks far from search/boresight while LOST.

4.4 Optional fine stage

When Link Readiness is high and residuals remain inside a capture cone for consecutive frames, control hands off to a mock 4-QD / FSM loop:

Parameter	Value
Enter readiness	$S \geq 0.70$
Exit readiness	$S \leq 0.55$ (hysteresis)
Tracking prerequisite	must be LOCKED
Capture cone enter	residual $\leq 600 \mu\text{rad}$
Capture drop	residual $> 900 \mu\text{rad}$
Stable frames	12 ($\sim 0.2\text{s}$ at 60 fps)
Fine slew authority	4000 $\mu\text{rad/s}$ class

This demonstrates coarse-to-fine handoff semantics beyond the mandatory coarse scope; it is a gated demo plant, not a calibrated flight FSM identification.

5 AI Methods

5.1 Motivation

Under Calm conditions, Classical centroid is sufficient. Under UAV/Stress grain, vibration, and scintillation, raw peak detection becomes brittle, while a learned beacon appearance model can retain lock. However, raw YOLO alone is fragile on out-of-distribution heavy grain; the production live AI path is therefore **hybrid fusion**.

5.2 Synthetic training

Frames are generated from the same scene and disturbance pipeline (`scripts/generate_training_data.py`), labeled in YOLO format for a single `beacon` class. A YOLOv8n detector is trained offline; weights load lazily at runtime (`weights/beacon_yolov8n.pt` or equivalent packaged weights). Training intentionally includes turbulence, vibration, and noise curricula so the detector sees the same channel family as the live loop.

5.3 Hybrid fusion pipeline (live AI path)

`HybridDetector` implements:

1. denoise / top-hat enhance the FOV image (`detector/preprocess.py`);
2. run YOLO on raw and enhanced views (weights 1.0 / 0.95);
3. run adaptive classical on enhanced and raw views (weights 0.92 / 0.88);
4. fuse by spatial consensus (~ 35 px agreement) or best weighted candidate;
5. apply a short temporal hold (8 frames) during detection gaps / re-acquisition.

For multi-beacon association, candidates are unioned and greedily NMS-pruned at the same consensus radius so two true beacons survive while duplicates collapse.

5.4 Classical vs AI comparison protocol

Matched-seed benchmarks (`scripts/run_comparison.py`) reseed disturbance RNG so Classical and AI observe identical channel realizations. Primary comparison metrics are lock retention and re-acquisition count. Mean pixel error is often slew-lag limited (~ 35 px even in Calm physics mode) and is reported honestly rather than oversold as a detector discriminator.

6 Test Methodology

6.1 Scenario presets

Table 3: Named disturbance scenarios used in demos and matrix runs

Scenario	Turb	Vib	Noise	Slew	Trajectory
Calm (F1)	0	0	0.05	90	Circular
UAV (F2)	4	8	0.2	90	Circular
Stress (F3)	7	15	0.35	120	Random walk

Additional motions include linear and figure-of-eight (PS Parameters requirement). Noise types (salt-pepper, Gaussian, Poisson) and weather grades (Clear / Haze / Fog / Rain / Low light) are selectable at runtime.

6.2 SIH evaluation layers

1. **Functional verification:** live GUI demo of mandatory features (10–15 min).
2. **Benchmark Performance-1:** execute scenarios; export centroid/performance logs.
3. **Benchmark Performance-2:** ingest judge-provided MP4 at ~ 30 fps with PTZ bypass (`scripts/benchmark_video.py`); report RMSE, acquisition/re-acquisition, lock retention, FPS.
4. **Technical evaluation:** architecture, algorithms, AI contribution, documentation, Q&A.

6.3 Automated metrics and tooling

Each run records simulation duration, FPS, acquisition time, average/maximum tracking error (pixel and μ rad), lock retention, re-acquisition statistics, and processing latency into CSV and JSON summaries. Streamlit (`report.py`) produces plots and a downloadable PDF performance report. Additional scientific checks:

- Acquisition Monte Carlo vs ESTOL-inspired $T_{\text{acq}} < 60$ s (`scripts/monte_carlo_acquisition.py`).
- Clutter Pd/Pfa scripts (`scripts/evaluate_detection.py`); Classical numbers are cited preferentially when AI JSON shows pathological false-alarm saturation.
- Ablation tooling (`scripts/run_ablation.py`) and demo capture (`scripts/record_demo_video.py`).

6.4 Parameters-table mapping

Table 4: Selected SIH26169 Parameters vs BeamLock

Parameter	Suggested	BeamLock
Screen size	$\geq 2000 \times 2000$	PS profile: 2000×2000
Camera resolution	640×480	PS profile: 640×480
Camera FOV	$4^\circ \times 3^\circ$ default	PS: $4^\circ \times 3^\circ$; Physics: 2.5 mrad class
Motions	≥ 4 incl. figure-8	Linear, circular, random, figure-8
Noise types	S&P, Gaussian, Poisson	Selectable (N)
Atmosphere grades	Clear–Low light	Weather presets (M)
Tracking error	≤ 10 px (guidance)	Offline B2 sample ~ 0.8 px; physics closed-loop slew-limited

7 Performance Analysis

7.1 Visual results

Figures 2a–2d show representative HUD captures under Calm, UAV, and Stress physics presets, plus re-acquisition after lock loss. Figures 3 show UAV locked tracking and recovery after a forced detector blind.

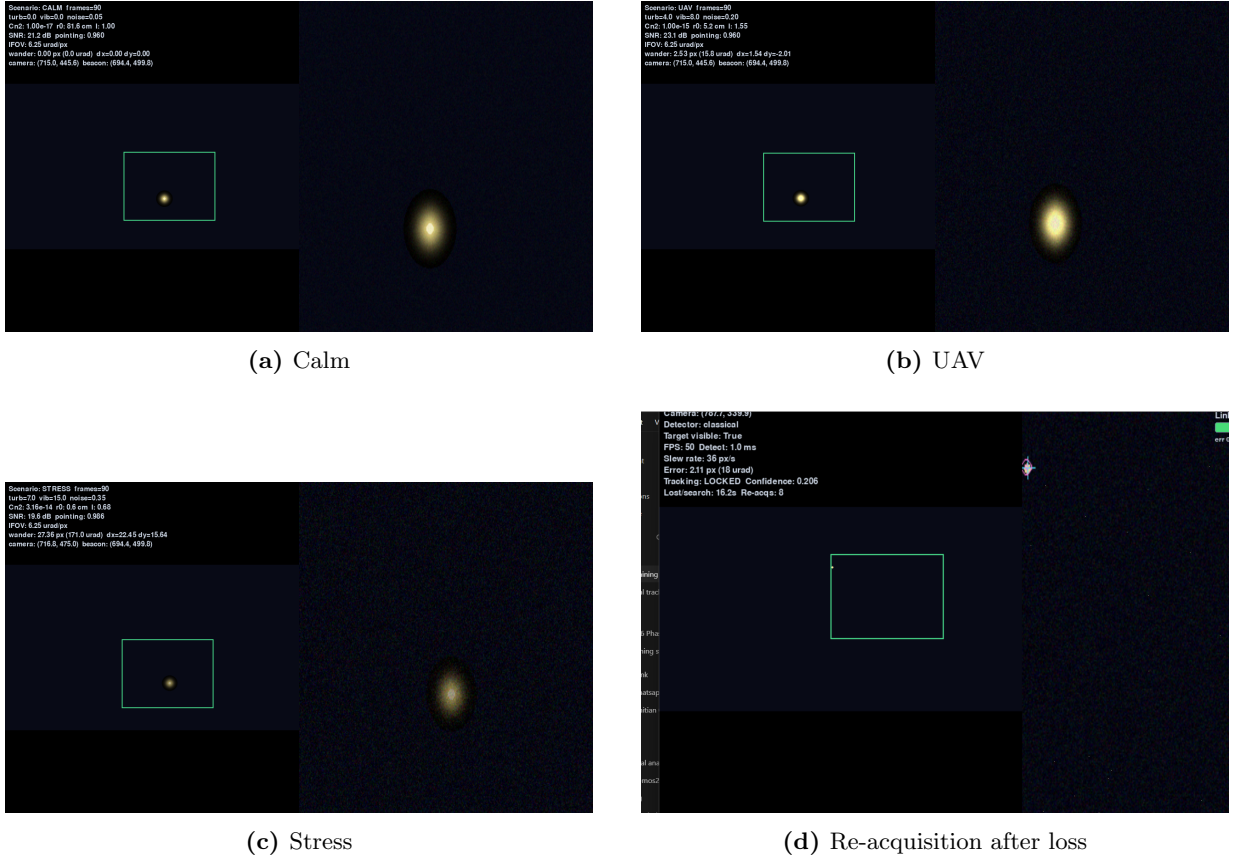


Figure 2: Live simulator captures (overview + camera FOV + HUD).

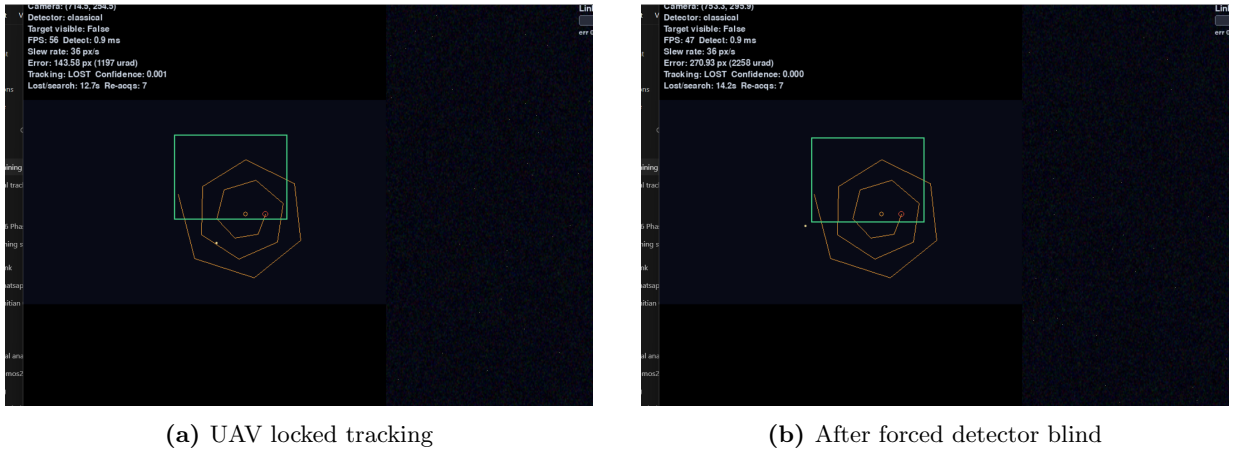


Figure 3: Additional operational captures used in demos and reports.

7.2 Matched-seed Classical vs AI matrix

Table 5: Scenario matrix (matched seed; mean error largely slew-lag limited)

Scenario	C err	AI err	C lock	AI lock	Reacq C/AI
Calm	35.5 px	35.5 px	100%	100%	1 / 1
UAV	35.6 px	35.6 px	97.7%	100%	1 / 1
Stress	35.1 px	36.1 px	95.5%	95.5%	2 / 1

Interpretation. AI (hybrid) improves lock stability under UAV and reduces re-acquisitions under Stress. Mean pixel error alone does not cleanly separate detectors because camera slew lag dominates the residual in physics mode ($\sim 35 \text{ px} \approx 219 \mu\text{rad}$ at $6.25 \mu\text{rad/px}$).

7.3 Acquisition Monte Carlo

Table 6: Time-to-lock Monte Carlo vs ESTOL-inspired 60 s bound

Scenario	Trials	Pass <60 s	Median T_{acq}	p95
UAV	40	40/40	0.82 s	0.85 s
Stress	40	40/40	0.97 s	1.0 s

7.4 Benchmark-2 sample (PTZ bypass)

On a synthetic full-frame noisy beacon MP4 with ground truth, Classical offline scoring achieved approximately **0.78 px average centroid error** and **0.86 px RMSE** with PTZ bypassed—well inside the suggested $\leq 10 \text{ px}$ tracking-error class for that offline path. This is the high-weight evaluation path when judges provide external video; closed-loop physics-mode mean error should not be conflated with Benchmark-2 centroid scoring.

7.5 Discussion

Performance should be judged primarily on lock retention, acquisition/re-acquisition timing, and reproducible logs. Physics-mode mean error near 35 px is a control-bandwidth effect, not necessarily a detection failure. PS-profile geometry and higher °/s slew improve the prospect of meeting the Parameters-table $\leq 10 \text{ px}$ guidance in closed-loop tests. Link Readiness and Fine handoff are evaluated qualitatively in demos (readiness rises under lock + SNR; Fine engages with hysteresis) rather than as a BER certificate.

7.6 Deliverables checklist

- Software application (Python entry + PyInstaller packaging script).
- Modular source code with documentation.
- This technical report (10–15 pages).
- User manual (installation, GUI, configuration).
- Automatic performance logs (CSV/JSON/Streamlit PDF).
- Optional demo video of the live application.

8 Future Improvements

1. Lengthen the optional demo video to a full 3–5 minutes with narration covering Calm $\rightarrow$ UAV $\rightarrow$ Stress $\rightarrow$ blind $\rightarrow$ Fine handoff.
2. Ship a pre-built standalone executable package alongside the PyInstaller script.

3. On-sky / real-camera fine-tuning of YOLO weights beyond synthetic data.
4. Longer Monte Carlo campaigns with confidence intervals and spiral-vs-raster A/B under matched seeds.
5. Optional offline phase-screen research mode while keeping the real-time statistical channel for demos.
6. Hardware-in-the-loop bridge: stream real camera frames into the same detect-track API used by Benchmark-2.
7. Closed-loop slew / prediction tuning in physics mode to shrink the ~ 35 px lag residual without destabilizing Stress re-acquisition.

9 Conclusion

BeamLock delivers an AI-assisted virtual coarse PAT system for SIH26169: configurable scene, multi-beacon support, AZ/EL camera control, Classical and hybrid AI detection, Kalman tracking, disturbance injection, real-time HUD, and automated performance reporting. Technical extensions—physics channel, Link Readiness Fine handoff, PS compliance profile, and MP4 Benchmark-2 tooling—strengthen evaluation readiness without abandoning the mandatory coarse-alignment scope. Verified evidence shows hybrid AI improving UAV lock retention and reducing Stress re-acquisitions, sub-second median acquisition under Monte Carlo, and sub-pixel offline Benchmark-2 centroid error with PTZ bypass. The project is positioned as a credible, reproducible software sandbox for PAT algorithm development under ISRO/DoS problem framing.

References

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End of SIH Technical Report — BeamLock / Team Cadets.